



SCEGGS Darlinghurst

2005
Preliminary Course
Semester 2 Examination

Name:

Teacher:

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1½ hours
- Write your name and your teacher's name at the top of each page
- Attempt **all** questions and show all necessary working
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used
- Begin each question on a new page
- **Do not** attach all questions together in one bundle

Total marks – 60

- Attempt Questions 1–5

Question	Total
1	/12
2	/12
3	/12
4	/12
5	/12
TOTAL	60
	%

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Instructions

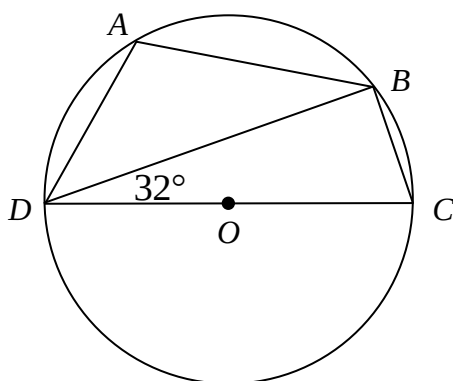
- Write your name and your teacher's name at the top of each page
 - Attempt **all** questions on the pad paper provided
 - Show all necessary working
 - Marks may be deducted for careless or badly arranged work
 - Mathematical templates, geometrical equipment and scientific calculators may be used
 - Begin each question on a new page
-

Question 1 (12 marks)**Marks**

- (a) A and B are the points $(1, 2)$ and $(5, 0)$ respectively. 2
Find the co-ordinates of the point P which divides AB in the ratio $3:1$.

- (b) A, B, C, D lie on the circumference of a circle, centre O . DC is the diameter and $\angle BDC = 32^\circ$. 2

Find $\angle BCD$ and $\angle BAD$ giving full reasons.



- (c) Consider the polynomial $P(x) = 4x^3 - 8x^2 - 3x + 9$.
- (i) Show that -1 is a zero of $P(x)$. 1
- (ii) Express $P(x)$ as a product of 3 linear factors. 2
- (iii) Hence, solve the inequality $P(x) \geq 0$. 2
- (d) Solve for x , 3

$$\frac{2}{x+3} \leq 5$$

Question 2 (12 marks)

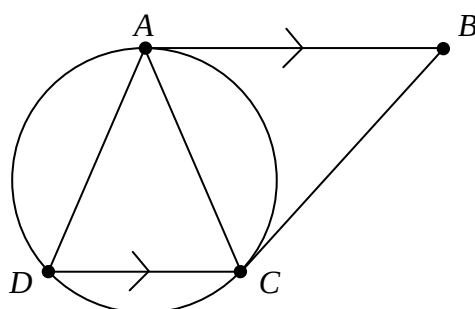
Marks

- (a) The polynomial $P(x) = x^3 + ax^2 + bx - 18$ has $x + 2$ as a factor and leaves a remainder of -24 when divided by $(x - 1)$. **3**

Find the values of a and b .

- (b) (i) Show that the equation of the tangent to the parabola $x = 2at, y = at^2$ at the point $t = p$ is given by $y = px - ap^2$. **2**
- (ii) Find the co-ordinates of the points where the tangent intersects the x and y axes. **2**
- (iii) Hence find the area of the triangle formed by the two intercepts and the origin. **1**

(c)



- A, C, D lie on the circumference of a circle. AB and CB are tangents at A and C respectively and AB is parallel to DC . **4**

Prove that $\triangle ABC \parallel \triangle DAC$.

Question 3 (12 marks)

Marks

(a) α, β and γ are roots of the equation $x^3 - 3x^2 + 4x + 8 = 0$

(i) Find the values of:

(1) $\alpha + \beta + \gamma$ 1

(2) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

(3) $\alpha^2 + \beta^2 + \gamma^2$ 2

(ii) Hence write down the cubic equation with roots $2\alpha, 2\beta$ and 2γ . 1

(b) (i) The equation of the chord of contact from the point $P(-1, -2)$ to the parabola $x^2 = 4y$ is given by $y = -\frac{x}{2} + 2$. Show that the points of contact are $(-4, 4)$ and $(2, 1)$. 2

(ii) Find the gradients of the tangents at these points of contact. 2

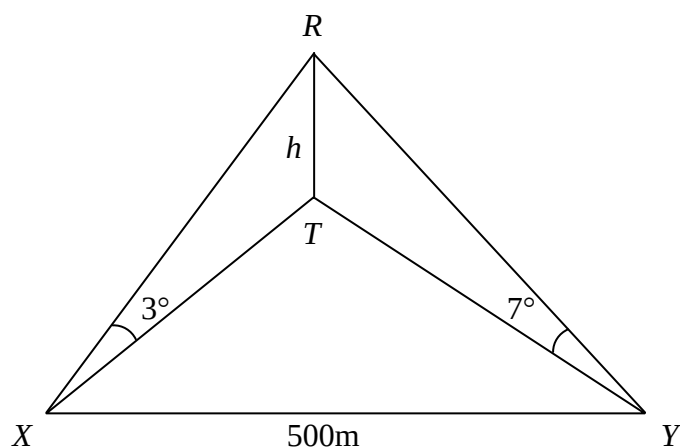
(iii) Hence determine the size of the acute angle formed between the tangents (to the nearest minute). 2

Question 4 (12 marks)		Marks
(a)	(i) Express $\cos x + \sin x$ in the form $R \cos(x - \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$.	2
	(ii) Hence solve $\cos x + \sin x = 1$ for $0^\circ < x \leq 360^\circ$.	2
(b)	(i) Write down the expansion of $\tan 2\theta$.	1
	(ii) Given that $t = \tan 22.5^\circ$, show that $t^2 + 2t - 1 = 0$.	2
	(iii) Hence show that $\tan 22.5^\circ = \sqrt{2} - 1$.	2
(c)	Show by substitution that the parametric equation	3
	$x = \sin \theta + \cos \theta$ $y = \tan \theta + \cot \theta$	
	satisfies the cartesian equation	
	$x^2 y = y + 2.$	

Question 5 (12 marks)

Marks

- (a) A knight in shining armour standing at point X observes the base of Rapunzel's tower (T) on a bearing of 035° and Rapunzel at the top of her tower (R) at an angle of elevation of 3° . The knight rides 500m to a point Y from which the tower is on a bearing of 300° and Rapunzel is at an angle of elevation of 7° .



- (i) Show that $\angle XTY = 95^\circ$. **1**

- (ii) Show that the height, h , of the tower (in metres) is given by **2**

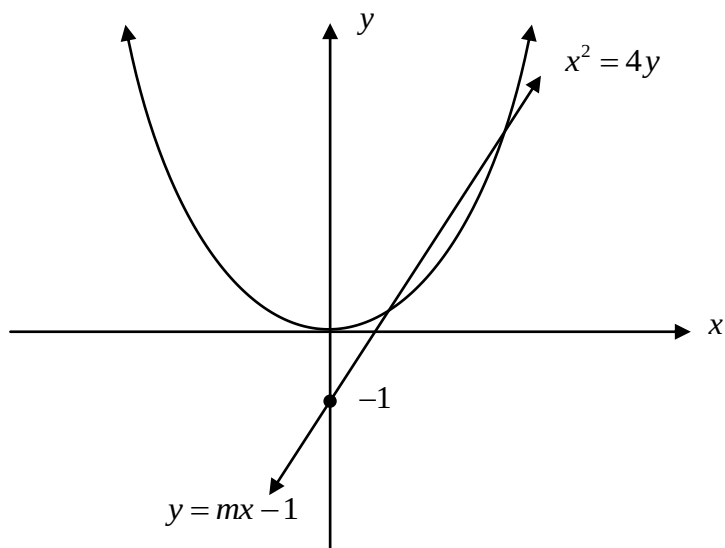
$$h = \frac{500}{\sqrt{\cot^2 3^\circ + \cot^2 7^\circ - 2 \cot 3^\circ \cot 7^\circ \cos 95^\circ}}$$

- (iii) How long must Rapunzel's hair be to reach the base of the tower? **1**

Question 5 continues on the next page

Question 5 (continued)

(b)



- (i) Show that the x co-ordinates of the points of intersection of the line $y = mx - 1$ and the parabola $x^2 = 4y$ satisfy the equation 1

$$x^2 - 4mx + 4 = 0$$

- (ii) For what values of m does the line $y = mx - 1$ and the parabola $x^2 = 4y$ have one or more points of intersection? 2

Question 5 continues on the next page

Question 5 (continued)

- (c) $P(2p, p^2)$ and $Q(2q, q^2)$ are variable points on the parabola $x^2 = 4y$ such that PQ passes through the point $(0, -1)$.

The equation of the chord PQ is given by $y = \left(\frac{p+q}{2} \right) x - pq$.

- (i) Show that $pq = 1$. **1**

- (ii) Show that the midpoint, M , of PQ is given by **1**

$$\left(p+q, \frac{p^2+q^2}{2} \right)$$

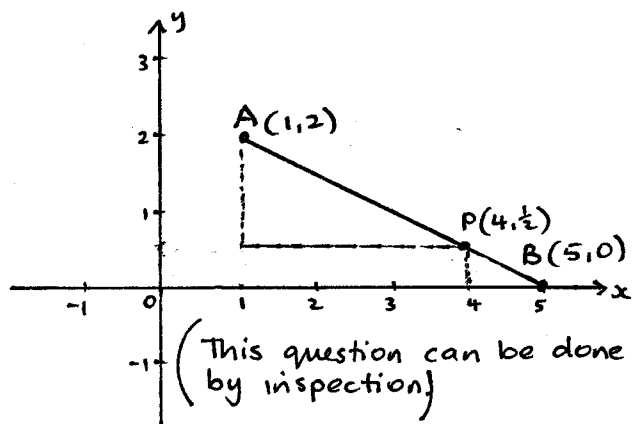
- (iii) Hence find the cartesian equation of the locus of M . **2**

- (iv) Using the result from part (b), explain why the domain of the locus of M is restricted to $x \geq 2$, $x \leq -2$. **1**

End of paper

Question 1

a)



$$\begin{array}{c} A(1,2) \quad B(5,0) \\ \swarrow \quad \searrow \\ 3:1 \end{array}$$

$$\begin{aligned} P &= \left(\frac{3 \times 5 + 1 \times 1}{3+1}, \frac{3 \times 0 + 1 \times 2}{3+1} \right) \\ &= \left(\frac{16}{4}, \frac{2}{4} \right) \\ &= \left(4, \frac{1}{2} \right) \end{aligned}$$

✓ ✓

The point $P(4, \frac{1}{2})$ divides AB in the ratio 3:1

• This question very well done.

b)

$$\angle DBC = 90^\circ \quad (\text{Angle in a semicircle is } 90^\circ \text{ at the circumf.})$$

$$\begin{aligned} \angle BCD &= 180^\circ - (90 + 32)^\circ \quad (\text{Angle sum } \triangle BDC = 180^\circ) \\ &= 58^\circ \end{aligned}$$

✓

$$\begin{aligned} \angle BAD &= 180^\circ - \angle BCD \quad (\text{opposite angles in a cyclic quad. are supplementary}) \\ &= 180^\circ - 58^\circ \\ &= 122^\circ \end{aligned}$$

✓

Since A, B, C, D lie on the circumference of the circle ABCD is a cyclic quadrilateral.

• Learn the correct wording of circle geometry reasons.

• Include this reason for angle sum of a triangle.

3 reasons → 2 marks.

c) $P(x) = 4x^3 - 8x^2 - 3x + 9$

i) $P(-1) = -4 - 8 + 3 + 9$
 $= 0$

$\therefore x = -1$ is a zero

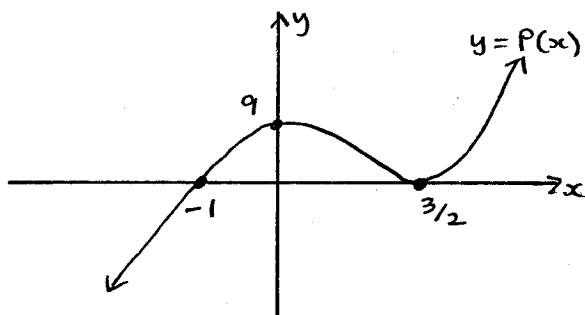
$\therefore (x+1)$ is a factor of $P(x)$

ii)

$$\begin{array}{r} 4x^2 - 12x + 9 \\ x+1 \overline{) 4x^3 - 8x^2 - 3x + 9} \\ \underline{4x^3 + 4x^2} \\ -12x^2 - 12x \\ \underline{-12x^2 - 12x} \\ 9x + 9 \\ \underline{9x + 9} \\ 0 \end{array}$$

$P(x) = (x+1)(4x^2 - 12x + 9)$ ✓
 $= (x+1)(2x-3)^2$ ✓

iii)



$P(x) \geq 0$

for $x \geq -1$

• This part was very well done.

Note $x = -1$ is a zero

$(x+1)$ is a factor

Since $(x+1)$ is a factor, there should be no remainder. If you have a remainder, something's gone wrong!

✓ one mark for correct sketch

✓ one mark for correct solution from your sketch.

d) $\frac{2}{x+3} \leq 5$ undefined for $x = -3$

multiply both sides by $(x+3)^2$

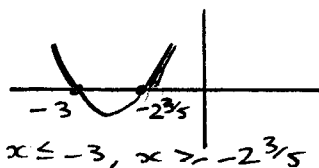
$\frac{2}{x+3} \times (x+3)^2 \leq 5(x+3)^2$

$2(x+3) \leq 5(x^2 + 6x + 9)$

$5x^2 + 30x + 45 - 2x - 6 \geq 0$

$5x^2 + 28x + 39 \geq 0$

$(5x + 13)(x + 3) \geq 0$ ✓



$x \leq -3, x \geq -2\frac{3}{5}$

$\therefore$ Solution ✓

$x < -3, x \geq -2\frac{3}{5}$ ✓

• Don't forget that the original is undefined when $x = -3$.

(This is an easy mark.)

{ ✓ one mark $x \neq -3$
 ✓ correct quadratic
 ✓ correct solution of your quadratic.

Question 2

Calc /2
Reas /4

(a) $(x+2)$ is a factor $\Rightarrow P(-2) = 0$

$$(-2)^3 + a(-2)^2 + b(-2) - 18 = 0$$

$$-8 + 4a - 2b - 18 = 0$$

$$4a - 2b = 26$$

$$\boxed{2a - b = 13} \quad (1)$$

$(x-1)$ leaves rem. of $-24 \Rightarrow P(1) = -24$

$$1^3 + a \cdot 1^2 + b \cdot 1 - 18 = -24$$

$$1 + a + b - 18 = -24$$

$$\boxed{a + b = -7} \quad (2)$$

Solve simult.

$$\begin{aligned} (1) + (2) &\Rightarrow 3a = 6 \\ &a = 2 \end{aligned}$$

$$\text{sub in } (2) \Rightarrow b = -9$$

(b) (i) $x = 2at$ $y = at^2$

$$P \text{ at } t = p \therefore \boxed{P(2ap, ap^2)}$$

$$\text{Gradient: } \frac{dx}{dt} = 2a \quad \frac{dy}{dt} = 2at$$

$$\frac{dy}{dx} = \frac{2at}{2a} = t$$

$$\therefore \boxed{m_t = p} \quad \checkmark \text{ Calc 2}$$

$$\begin{aligned} \text{Eqn tang: } y - ap^2 &= p(x - 2ap) \checkmark \\ y - ap^2 &= px - 2ap^2 \\ y &= px - ap^2 \end{aligned}$$

(ii) x int: $y = 0$

$$0 = px - ap^2$$

$$x = ap$$

$$(ap, 0) \checkmark$$

y int: $x = 0$

$$y = p \cdot 0 - ap^2$$

$$y = -ap^2$$

$$(0, -ap^2) \checkmark$$

• Generally people knew the factor/rem. theorems or they didn't.

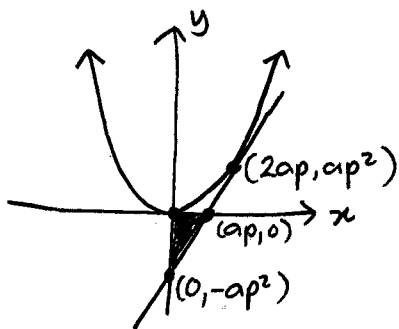
If you are someone who doesn't know it - learn it!

• You needed to show the gradient of the tangent = p by differentiating first.

• If there are no t s in the question & you know $t = p$, you should substitute!

• You could have answered from (ii) $\rightarrow$ even if you didn't understand (i). You need to be able to recognise when you can do this.

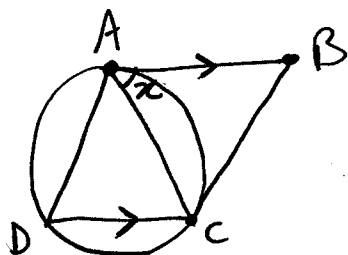
(iii)



$$\begin{aligned}\text{Area } \Delta &= \left| \frac{1}{2} \times \text{base} \times \text{height} \right| \\ &= \left| \frac{1}{2} \times ap \times ap^2 \right| \\ &= \left| \frac{a^2 p^3}{2} \right| \quad \checkmark\end{aligned}$$

• Technically you need the absolute value signs for the area. 'p' can certainly be -ve.

(c)



$$\text{Let } \angle BAC = x$$

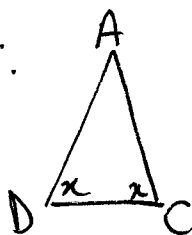
$$AB = BC \text{ (tang. from an ext. pt. are =)} \quad \checkmark$$

$$\therefore \Delta ABC \text{ is isos.}$$

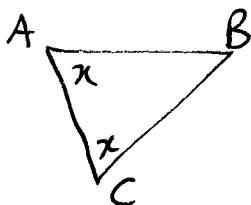
$$\therefore \angle BAC = x = \angle BCA \text{ (Ls opp. sides of isos } \Delta =) \quad \checkmark$$

$$\angle BAC = x = \angle ACD \text{ (alt. Ls on // lines AB \& DC =)} \quad \checkmark$$

$$\angle BAC = x = \angle ADC \text{ (L bet. tang. \& chord = L in alt. seg.)} \quad \checkmark$$



&



Reas.
4

$$\therefore \Delta ABC \equiv \Delta DAC \text{ (equiangular)}$$

• There are various ways to prove this question

You need to be careful with the alternate segment theorem eg. $\angle BCA \neq \angle DAC$!

You also need to be careful when naming angles - one incorrect letter can mean a world of difference.

Question 3

a) $x^3 - 3x^2 + 4x + 8 = 0$

i) $\alpha + \beta + \gamma = -b/a$
 $= -(-3)/1$
 $= 3$ ✓

ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$ ✓
 $= \frac{\frac{c}{a}}{-d/a}$
 $= \frac{4/1}{-8/1}$
 $= -\frac{1}{2}$ ✓

(3) $\alpha^2 + \beta^2 + \gamma^2$

Note $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 3^2 - 2 \times 4$ ✓
 $= 9 - 8$
 $= 1$ ✓

ii) The equation has the form
 $ax^3 + bx^2 + cx + d = 0$
 $x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a}x$

Sum of roots

$2\alpha + 2\beta + 2\gamma = -\frac{b}{a}$
 $2(\alpha + \beta + \gamma) = -\frac{b}{a}$
 $2 \times 3 = -\frac{b}{a}$
 $\therefore \frac{b}{a} = -6$

Sum two at a time

$2\alpha \cdot 2\beta + 2\alpha \cdot 2\gamma + 2\beta \cdot 2\gamma = \frac{c}{a}$
 $4(\alpha\beta + \alpha\gamma + \beta\gamma) = \frac{c}{a}$
 $4 \times 4 = \frac{c}{a}$
 $\therefore \frac{c}{a} = 16$

Product.

$2\alpha \cdot 2\beta \cdot 2\gamma = -\frac{d}{a}$
 $8\alpha\beta\gamma = -\frac{d}{a}$
 $8 \times -8 = -\frac{d}{a}$
 $\therefore \frac{d}{a} = 64$

✓
Reas

$\therefore$ Equation of cubic equation

$x^3 - 6x^2 + 16x + 64 = 0$

LEARN THE RULES!!!

$ax^3 + bx^2 + cx + d = 0$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$
$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$
$$\alpha\beta\gamma = -\frac{d}{a}$$

This question was well done as almost everyone knew their rules. ii

These results came from part(a)

$$\left. \begin{aligned} \alpha + \beta + \gamma &= -\frac{b}{a} = 3 \\ \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} = 4 \\ \alpha\beta\gamma &= -\frac{d}{a} = -8 \end{aligned} \right\} \begin{array}{l} \text{use these values} \\ \text{in part(ii)} \end{array}$$

• Another method is to write the equation as

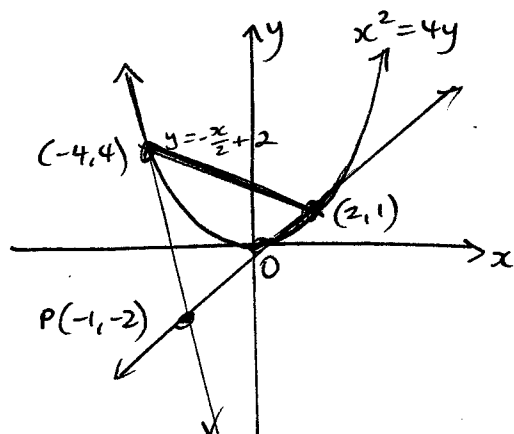
$(x - 2\alpha)(x - 2\beta)(x - 2\gamma) = 0$

then expand and collect like terms.
then substitute the values for sum, product etc. found in part(a).

• part(ii) was a bit tricky.

It had to be perfectly correct to gain one mark.

b) i)



$$\begin{cases} \textcircled{1} y = -\frac{x}{2} + 2 \\ \textcircled{2} x^2 = 4y \end{cases} \text{ Solve simultaneously}$$

substitute $\textcircled{1}$ into $\textcircled{2}$

$$x^2 = 4\left(-\frac{x}{2} + 2\right)$$

$$x^2 = -2x + 8$$

$$x^2 + 2x - 8 = 0$$

$$(x+4)(x-2) = 0$$

$$x = -4$$

$$x = 2$$

$$y = \frac{-4}{2} + 2$$

$$y = 4$$

$$y = \frac{-1+2}{2}$$

$$y = 1$$

$$(-4, 4)$$

$$(2, 1)$$

points of contact

$$\begin{aligned} \text{ii)} \quad x^2 &= 4y \\ y &= \frac{x^2}{4} \\ \frac{dy}{dx} &= \frac{2x}{4} \\ &= \frac{x}{2} \end{aligned}$$

At $(-4, 4)$ gradient tangent

$$m_1 = -\frac{4}{2} = -2$$

At $(2, 1)$ gradient tangent

$$m_2 = \frac{2}{2} = 1$$

Calc

$$\begin{aligned} \text{iii)} \quad \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{-2 - 1}{1 - 2 \times 1} \right| \\ &= \left| \frac{-3}{-1} \right| \\ &= 3 \end{aligned}$$

$$\therefore \theta = 71^\circ 34'$$

(to nearest minute)

• It is not sufficient to substitute the points $(-4, 4)$ and $(2, 1)$ into $y = -\frac{x}{2} + 2$. This shows only that the points satisfy this line's equation.

• Another method is to find gradients using two end points

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

If you're going to use a formula make sure you know it and you get the signs correct !!!!!

Question 4

Reas. / 7

(a) (i) $\cos x + \sin x$

$$R \cos(x - \alpha)$$

$$= R \cos x \cos \alpha + R \sin x \sin \alpha$$

$$\therefore \begin{matrix} R \cos \alpha = 1 & \textcircled{1} \\ R \sin \alpha = 1 & \textcircled{2} \end{matrix} \left. \vphantom{\begin{matrix} R \cos \alpha = 1 \\ R \sin \alpha = 1 \end{matrix}} \right\} \text{Solve} \quad \checkmark$$

$$\frac{\textcircled{2}}{\textcircled{1}} \Rightarrow \tan \alpha = 1$$

$$\alpha = 45^\circ$$

$$\textcircled{1}^2 + \textcircled{2}^2 \Rightarrow R = \sqrt{2} \quad \checkmark$$

$$\therefore \cos x + \sin x = \sqrt{2} \cos(x - 45^\circ)$$

(ii) Instead, solve

$$\sqrt{2} \cos(x - 45^\circ) = 1$$

$$\cos(x - 45^\circ) = \frac{1}{\sqrt{2}}$$

$$\begin{matrix} \text{S} & \text{A} \\ \text{---} & \text{---} \\ \text{T} & \text{C} \end{matrix} \checkmark \quad (x - 45^\circ) = 45^\circ, 315^\circ$$

$$x = 90^\circ, 360^\circ \quad \checkmark \quad \checkmark$$

(b) (i) $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} \quad \checkmark$

(ii) let $\theta = 22.5^\circ$ above

$$\Rightarrow \tan 45^\circ = \frac{2 \tan 22.5^\circ}{1 - \tan^2 22.5^\circ} \quad \checkmark$$

$$1 = \frac{2t}{1 - t^2}$$

$$1 - t^2 = 2t \quad \checkmark$$

$$t^2 + 2t - 1 = 0$$

(iii) $t = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times -1}}{2 \times 1} \quad \text{Reas. 4}$

$$= \frac{-2 \pm \sqrt{8}}{2} = -1 \pm \sqrt{2} \quad \checkmark$$

$\therefore \tan 22.5^\circ = -1 + \sqrt{2}$ or $-1 - \sqrt{2}$
since 22.5° is in 1st quad. $\checkmark$
 $\tan 22.5^\circ = -1 + \sqrt{2}$

• Auxiliary angle method is a standard method for solving trig equations — & one you need to know!

• Surprising is how many of you knew how to solve the equation⁽ⁱⁱ⁾ but didn't understand how to answer part (i).

• Watch the domain — there was actually no trick this time, but people assumed there was. $0^\circ < x \leq 360^\circ$.

(i) is a GIVEAWAY! A ridiculous # of people got it wrong.

• They wouldn't give you such an ^{easy} mark for part (i) unless they wanted you to use it. So use it!

• (iii) is doable even if you got lost previously. When you get lost you must scan down to see where you can pick up the question again.

(c) $x = \sin \theta + \cos \theta$
 $y = \tan \theta + \cot \theta$

Prove $x^2 y = y + 2$

$$\begin{aligned} \text{LHS} &= (\sin \theta + \cos \theta)^2 \times y \\ &= (\sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta) y \\ &= (1 + 2 \sin \theta \cos \theta) y \\ &= y + 2 \sin \theta \cos \theta y \\ &= y + 2 \sin \theta \cos \theta \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right) \\ &= y + 2 \sin^2 \theta + 2 \cos^2 \theta \\ &= y + 2(\sin^2 \theta + \cos^2 \theta) \\ &= y + 2 \\ &= \text{RHS} \end{aligned}$$

Reas. 3

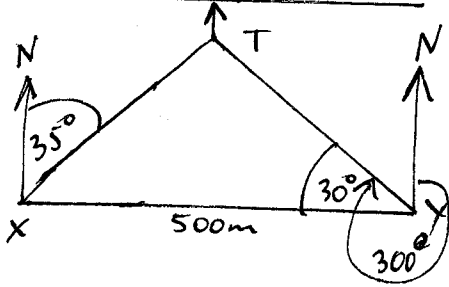
• $x^2 \neq \sin^2 \theta + \cos^2 \theta$!

$$\begin{aligned} x^2 &= (\sin \theta + \cos \theta)^2 \\ &= \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta \end{aligned}$$

• Writing up such proofs are easier if you use LHS / RHS - type setting out.

Question 5

a) Birds Eye view



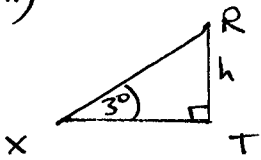
$$i) \angle XTY = 300^\circ - 270^\circ = 30^\circ$$

$$\angle TXY = 90^\circ - 35^\circ = 55^\circ \quad (\text{right } \angle = 90^\circ)$$

$$\therefore \angle XTY = 180^\circ - (30^\circ + 55^\circ) \quad (\angle \text{sum } \Delta = 180^\circ) = 95^\circ$$

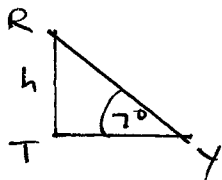
You should list out steps to show why $\angle XTY = 95^\circ$. Although it is not a full geometric proof it is still not sufficient to just write angles on a diagram with no clear reasoning.

ii)



$$\tan 30^\circ = \frac{h}{XT}$$

$$XT = \frac{h}{\tan 30^\circ} = h \cot 30^\circ$$



$$\tan 70^\circ = \frac{h}{TY}$$

$$TY = \frac{h}{\tan 70^\circ} = h \cot 70^\circ$$

- ① mark for expressions for both XT and TY
- ① mark for use of the cosine rule to find h → showing clear factorising step. (no fudging!)

Using the cosine rule in ΔXTY

$$500^2 = XT^2 + TY^2 - 2XT \cdot TY \cdot \cos 95^\circ$$

$$250000 = (h \cot 30^\circ)^2 + (h \cot 70^\circ)^2 - 2h \cot 30^\circ \cot 70^\circ \cos 95^\circ$$

(-factorise h^2)

$$500^2 = h^2 (\cot^2 30^\circ + \cot^2 70^\circ - 2 \cot 30^\circ \cot 70^\circ \cos 95^\circ)$$

$$\therefore h^2 = \frac{500^2}{\cot^2 30^\circ + \cot^2 70^\circ - 2 \cot 30^\circ \cot 70^\circ \cos 95^\circ}$$

$$h = \frac{500}{\sqrt{\cot^2 30^\circ + \cot^2 70^\circ - 2 \cot 30^\circ \cot 70^\circ \cos 95^\circ}}$$

$$iii) \hat{=} 23.38 \text{ m}$$

Calculator step.

Worth practising! (1 mark)

b) i) Solve simultaneously

$$y = mx - 1 \quad (1)$$

$$x^2 = 4y \quad (2)$$

Substitute (1) into (2)

$$x^2 = 4(mx - 1)$$

$$x^2 = 4mx - 4$$

$$x^2 - 4mx + 4 = 0 \quad \checkmark$$

ii) One or more points of intersection means that this quadratic equation has one or more solutions.
 $\therefore \Delta \geq 0$

$$\Delta = b^2 - 4ac$$

$$= (-4m)^2 - 4 \cdot 1 \cdot 4$$

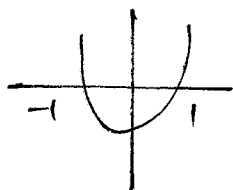
$$= 16m^2 - 16$$

Solve $\Delta \geq 0$

$$16m^2 - 16 \geq 0$$

$$16(m^2 - 1) \geq 0$$

$$16(m-1)(m+1) \geq 0$$



$$m \leq -1, m \geq 1 \quad \checkmark$$

For these values of the gradient, the line and the parabola have one or more points of intersection.

Reas 2

most students managed this part of part b).

Applications of the Discriminant

For points of intersection between line and a parabola.

Solve simultaneously and then apply to the quadratic equation formed.

$\Delta \geq 0$ one or more solutions

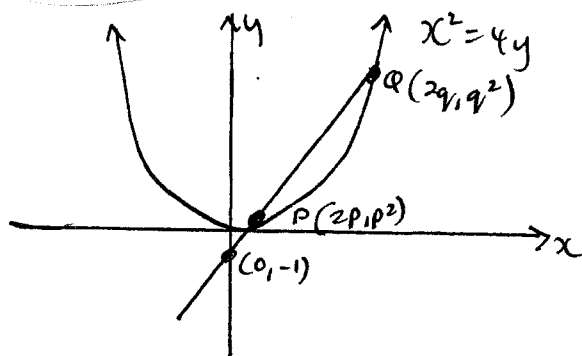
$\Delta > 0$ more than one solution

$\Delta = 0$ one solution $\rightarrow$ line is a tangent

$\Delta < 0$ no solutions
 $\rightarrow$ line and curve have no intersection.

It is always a good idea to be aware of these applications. The use of the discriminant is a technique that appears frequently in harder questions in both the 2U and Ext 1 exams.

c)



Chord PQ, $y = \left(\frac{p+q}{2}\right)x - pq$

Draw a sketch.
It helps sort out the logic in this question.

i) Chord PQ passes through the point $(0, -1)$

Substitute this point into PQ

$$y = \left(\frac{p+q}{2}\right)x - pq$$

$$-1 = 0 - pq$$

$$\therefore pq = 1$$



Reas 1

ii) $P(2p, p^2)$ $Q(2q, q^2)$

midpoint $M = \left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right)$

$$= (p+q, \frac{p^2+q^2}{2})$$



• Easy one mark here.
• Very well done by everyone who attempted it.

iii) Locus of M

$$x = p+q$$

$$y = \frac{1}{2}(p^2+q^2)$$

$$= \frac{1}{2}((p+q)^2 - 2pq)$$

$$= \frac{1}{2}(x^2 - 2 \times 1)$$

$$= \frac{1}{2}(x^2 - 2)$$



$$2y = x^2 - 2$$

$$x^2 = 2y + 2$$

$$x^2 = 2(y+1)$$

$$x^2 = 4 \cdot \frac{1}{2}(y+1)$$

The locus of M is a parabola.

learn this

$$p^2+q^2 = (p+q)^2 - 2pq.$$

This substitution is very common in locus problems.

Reas 2

iv) See next page for 3 different

Solutions that were presented by students.

Well done if you were responsible for any of them! v

①

iv) Since the locus of M is formed by a chord on the parabola $x^2 = 4y$, this chord has one or more points of intersection when $m \leq -1, m \geq 1$ (from part(b))

Since the chord has equation

$$y = \left(\frac{p+q}{2}\right)x - pq$$

(substitute $pq = 1$)

$$= \left(\frac{p+q}{2}\right)x - 1$$

this matches the line's equation given in part (b)

$$y = mx - 1$$

matching the gradients

$$\frac{p+q}{2} = m$$

$$m \leq -1, m \geq 1$$

$$\frac{p+q}{2} \leq -1, \frac{p+q}{2} \geq 1$$

$$p+q \leq -2, p+q \geq 2$$

Since $x = p+q$ for point M

$$x \leq -2, x \geq 2$$

✓ (Reason)

②

Outside the parabola, there is no chord and hence no midpoint.

The chord only exists inside or on the parabola $x^2 = 4y$.

The region inside the parabola is described by $x^2 \leq 4y$

the locus of M

$$x^2 = 2y + 2$$

$$2y = x^2 - 2$$

substitute this locus into the region

$$x^2 \leq 4y$$

$$x^2 \leq 2 \cdot 2y$$

$$x^2 \leq 2 \cdot (x^2 - 2)$$

$$x^2 \leq 2x^2 - 4$$

$$-x^2 \leq -4$$

$$x^2 \geq 4$$

$$(x-2)(x+2) \geq 0$$

$$\frac{x-2}{x+2}$$

$$x \leq -2, x \geq 2$$

The locus of M lies within the parabola for this domain.

③

parabola $x^2 = 4y, y = \frac{x^2}{4}$

chord $y = \left(\frac{p+q}{2}\right)x - 1$

points of intersection

solve simultaneously

$$\frac{x^2}{4} = \left(\frac{p+q}{2}\right)x - 1$$

$$x^2 = 2(p+q)x - 4$$

$$x^2 - 2(p+q)x + 4 = 0$$

$$\Delta = b^2 - 4ac$$

$$= (2(p+q))^2 - 4 \cdot 1 \cdot 4$$

$$= 4(p+q)^2 - 16$$

for the chord and parabola to have one or more points of intersection

$$\Delta \geq 0$$

$$4(p+q)^2 - 16 \geq 0$$

$$(p+q)^2 - 4 \geq 0$$

$$(p+q-2)(p+q+2) \geq 0$$

for the locus of M $x = p+q$

$$(x-2)(x+2) \geq 0$$

$$\frac{x-2}{x+2}$$

$$x \leq -2, x \geq 2.$$

Note: This method is using the same logic as in part(b) just a bit more complicated algebra.